

Oil Price Uncertainty in an Era of Conflict

Who Pays More? Europe vs Asia Under the Shadow of the Iran Crisis

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1 Introduction

The global oil market is one of the most geopolitically sensitive commodity markets in the world. Price movements are rarely driven by supply and demand alone — they respond sharply to conflict, sanctions, and the threat of supply disruption. The escalating tensions involving Iran in 2024–2025, including US-Israel military strikes and fears over the Strait of Hormuz, have reintroduced a level of uncertainty not seen since the 2022 energy crisis.

This report investigates a deceptively simple question: does every region of the world suffer equally when oil prices spike? The answer is no — and the divergence between European and Asian oil benchmarks is the evidence.

We analyse two benchmark prices:

- **Brent Crude** (DCOILBRENTU): the European benchmark, traded on the ICE exchange, priced in USD per barrel.
- **Dubai Crude** (POILDUBUSDM): the Asian benchmark used to price roughly two-thirds of Middle Eastern exports to Asia, including Japan, China, South Korea, and India.

Both series are retrieved directly from the FRED database (Federal Reserve Bank of St. Louis) using the official API — no manual CSV download required.

2 Data

Table 1: Descriptive statistics — Brent (Europe) vs Dubai (Asia), 2010–present. All prices in USD/barrel.

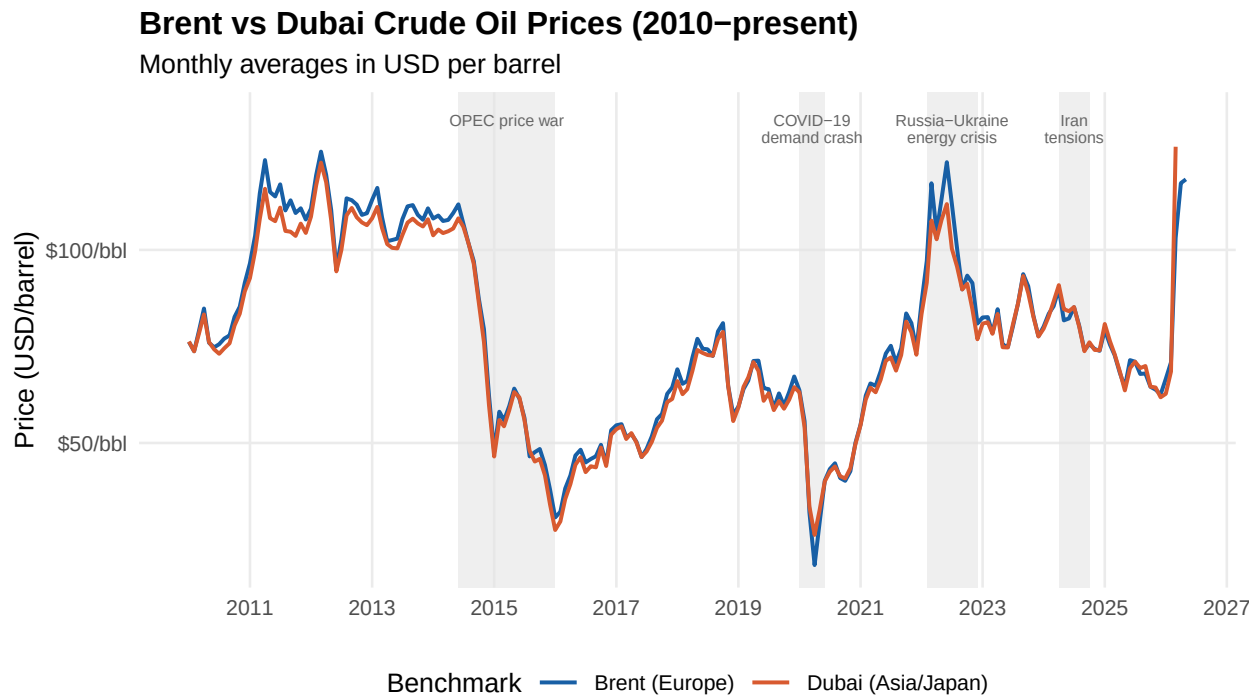
Benchmark	N	USD per barrel				
		Mean	Median	Min	Max	Std. Dev.
Brent (Europe)	197	78.01	75.44	18.38	125.45	24.18
Dubai (Asia/Japan)	195	75.91	74.09	26.19	126.71	23.00

The dataset covers monthly oil prices from January 2010 to the present. Brent is originally reported at daily frequency and aggregated to monthly averages for comparability with the Dubai series, which is reported monthly by the IMF. The sample yields approximately 197 monthly observations per benchmark — sufficient for robust time series estimation, as forecasting models require trend identification rather than large cross-sectional samples.

The mean price over the sample is \$78.01/bbl for Brent and \$75.91/bbl for Dubai, with Brent trading on average \$2.10/barrel above Dubai. However, this average masks substantial variation: the spread has ranged from +\$11.61/bbl (Europe paying more) to −\$23.57/bbl (Asia paying a severe premium), reflecting the asymmetric impact of geopolitical shocks on regional supply routes.

3 Price Dynamics: Europe vs Asia

3.1 Historical Price Trajectories



Source: FRED (EIA / IMF). Brent = DCOILBRETEU, Dubai = POILDUBUSDM. Author: Daniela Borja Polanco.

Figure 1: Monthly oil prices: Brent (Europe) vs Dubai (Asia/Japan), 2010–present. Shaded areas mark key geopolitical events. Source: FRED (EIA / IMF). Author: Daniela Borja Polanco.

The chart reveals the fundamental structure of the global oil market: **Brent and Dubai move almost identically most of the time**, confirming that oil is an integrated global commodity. When supply or demand shocks occur anywhere, both benchmarks respond together. The OPEC price war (2014–2016) collapsed prices from \$110 to \$30/barrel across both benchmarks simultaneously. The COVID-19 crash (2020) sent both to historic lows in a matter of weeks.

The **interesting economics lies in the moments of divergence** — when one benchmark decouples from the other. These divergences, invisible in Figure 1 but clearly visible in the spread analysis below, reveal which regions bear the greater cost of geopolitical disruption. The sharp spike in both series at the far right (late 2025 / early 2026) corresponds to the most recent Iran escalation — and notably, the spread chart shows that Dubai spiked *more* than Brent during this episode, meaning Asia bore a disproportionate share of the price shock.

3.2 The Brent–Dubai Spread

The spread (Brent minus Dubai) is the key analytical instrument of this report.

Positive spread (Brent > Dubai): Europe pays more per barrel than Asia.

Negative spread (Dubai > Brent): Asia pays more per barrel than Europe.

Historically Brent trades slightly above Dubai, but this relationship breaks down during geopolitical shocks — particularly those threatening Middle Eastern supply routes like the Strait of Hormuz.

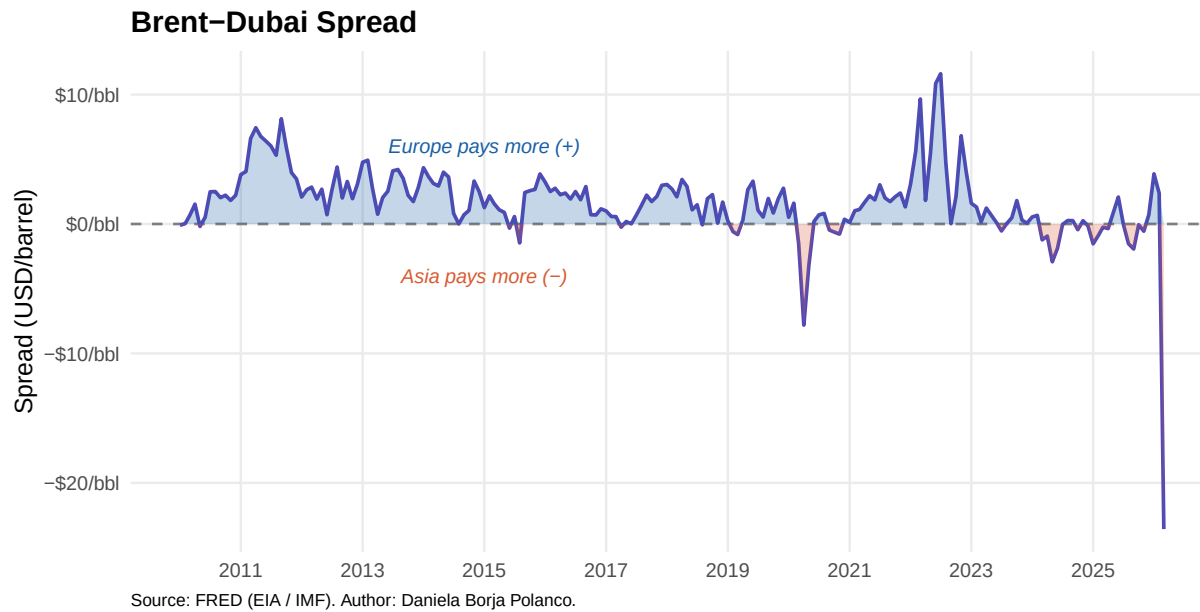


Figure 2: Brent-Dubai spread. Positive = Europe pays more. Negative = Asia pays more.

The spread chart tells a more nuanced story than the price chart alone. For most of 2010–2024, Europe paid a modest premium — consistent with Brent’s historical role as the global reference price. The spread was narrow and relatively stable, rarely exceeding \$5/bbl in either direction.

The two most dramatic exceptions are revealing. First, a brief episode around 2020 where the spread turned sharply negative — Dubai temporarily priced above Brent — as COVID-19 disrupted Asian demand patterns unevenly. Second, and far more striking, the collapse at the right edge of the chart: in late 2025/early 2026 the spread plunged to nearly **–\$23/barrel**, meaning Asia was paying over \$23 more per barrel than Europe. This is not a marginal difference. At Japan’s import volume of roughly 3 million barrels per day, a \$23 spread sustained over a year represents approximately **\$25 billion in additional annual energy costs** relative to European competitors — a macroeconomic shock, not a statistical artefact.

4 Iran, the Strait of Hormuz, and Regional Exposure

4.1 Why Geography Determines Pain

The Strait of Hormuz — the narrow waterway between Iran and Oman — is the most critical oil chokepoint in the world. Approximately 20% of global oil trade passes through it daily. Any military escalation involving Iran carries the risk of partial or full closure of the Strait, with dramatically asymmetric consequences by region:

Table 2: Regional oil import dependence and Strait of Hormuz exposure.
Source: IEA World Energy Balances; BP Statistical Review of World Energy. Author: Daniela Borja Polanco.

Region	Benchmark	ME share of imports (%)	Hormuz exposure	Alternative supply routes
Japan	Dubai	90	Critical	Very limited
South Korea	Dubai	70	Critical	Very limited
China	Dubai/Oman	45	High	Russia/pipeline
India	Dubai	65	High	Russia/Africa
Germany	Brent	30	Moderate	Norway/Russia
France	Brent	25	Moderate	Norway/Algeria
United States	WTI	10	Low	Canada/domestic

The table reveals a clear vulnerability gradient. Japan and South Korea sit at the top of the risk ladder: both source the overwhelming majority of their oil from the Middle East, and neither has a realistic alternative at scale. China and India occupy an intermediate position — high dependence, but with meaningful alternatives through Russian supply that have become more actively used since 2022. European economies, priced on Brent, draw primarily from the North Sea and North Africa, with the Middle East representing a smaller and increasingly substitutable share of supply.

4.2 Forecasting Price Paths Under Uncertainty

We use Simple Exponential Smoothing (SES) and Holt’s damped trend method to generate 12-month forecasts for both benchmarks. The confidence intervals serve as a quantitative measure of price risk — a wider band signals a more unpredictable environment.

Table 3: Model performance — SES vs Holt (damped trend).
Source: FRED (EIA / IMF). Author: Daniela Borja Polanco.

Benchmark	Model	Alpha	RMSE
Brent (Europe)	SES	0.9999	6.555
Brent (Europe)	Holt (damped trend)	NA	6.503
Dubai (Asia/Japan)	SES	0.9999	6.773
Dubai (Asia/Japan)	Holt (damped trend)	NA	6.702

Both models return $\alpha = 0.9999$ for both benchmarks — effectively $\alpha = 1$. This means the optimal forecast at every point is almost entirely determined by the most recent price observation, with historical prices carrying negligible weight. This is the statistical signature of a **random walk**: oil prices are shock-driven rather than trend-driven, and the best prediction of tomorrow’s price is today’s price. This finding is well-established in the commodity economics literature and is consistent with the efficient markets hypothesis applied to oil futures.

The marginal improvement of Holt over SES is minimal — less than 1% lower RMSE for both benchmarks — confirming that adding a trend component provides almost no additional forecasting power. Oil does not trend predictably; it jumps in response to events.

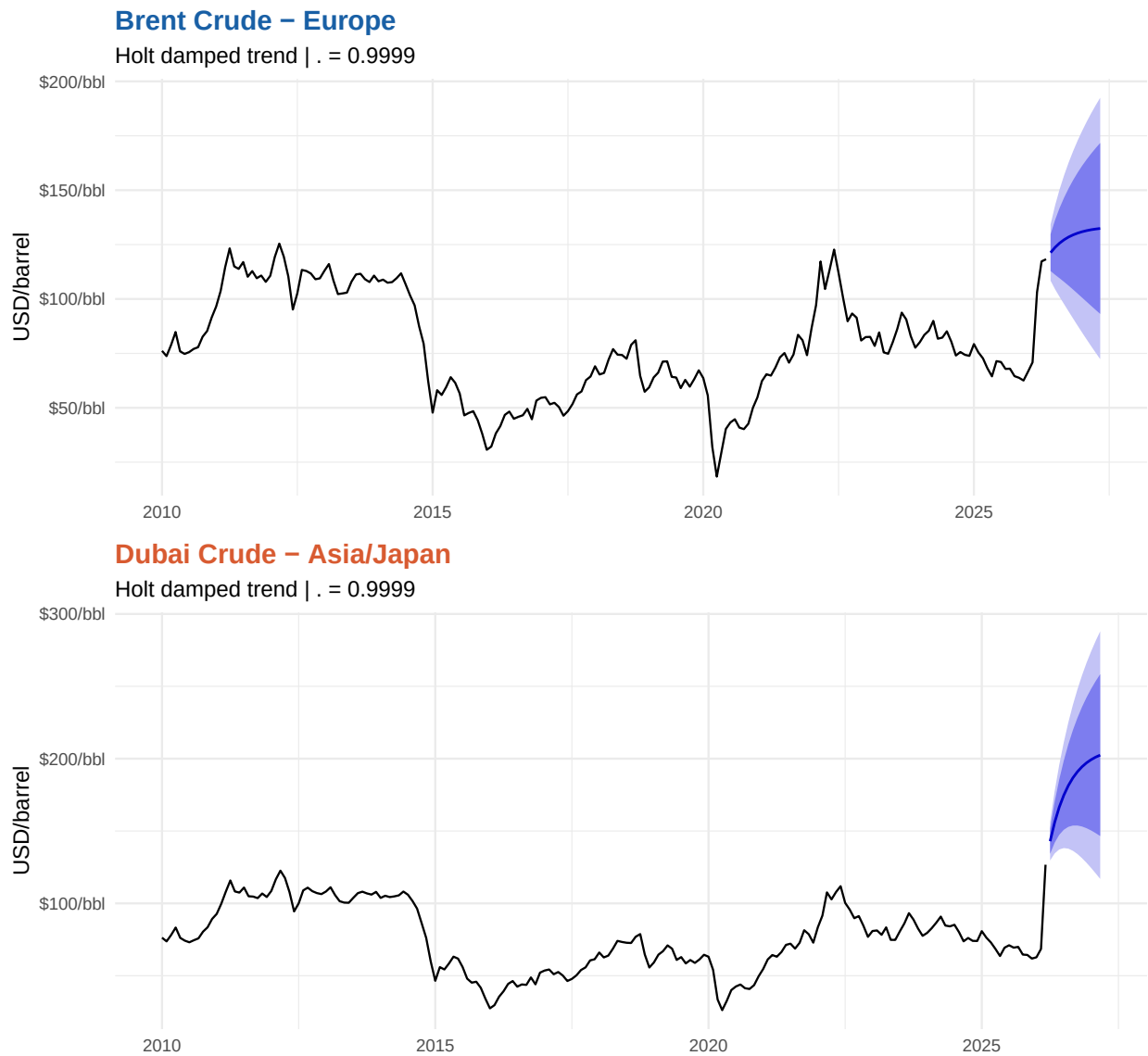


Figure 3: 12-month forecasts (Holt damped trend) for Brent and Dubai crude. Shaded bands = 80% and 95% prediction intervals. Source: FRED (EIA / IMF). Author: Daniela Borja Polanco.

Interpreting the forecasts. The Brent forecast (top panel) projects prices around \$130/barrel with a wide but plausible confidence band — reflecting genuine market uncertainty, but anchored to a realistic price range given the recent spike.

The Dubai forecast (bottom panel) requires more careful interpretation. The model projects prices above \$200/barrel, with an upper confidence bound approaching \$300/barrel. This is not a credible point forecast — it is a direct consequence of $\alpha = 0.9999$ combined with an extreme upward spike in the final observation. The model sees a sharp recent upward movement and, placing almost all weight on that last data point, extrapolates it mechanically forward.

This should be read not as a price prediction but as a **quantification of extreme uncertainty**. The

confidence interval is so wide it is essentially saying: *we do not know*. And that, in itself, is the most important finding of the forecasting exercise. Asian crude markets have become genuinely unforecastable under current geopolitical conditions. A mechanical time series model cannot distinguish between a temporary geopolitical spike and a structural regime change — and neither, at this moment, can the market.

Who bears the greatest uncertainty? Japan and South Korea face not only higher expected prices but far greater price variance. With $\alpha \approx 1$, every geopolitical shock lands directly in the price with no dampening from historical averages. For an economy like Japan — where 90% of oil imports transit the Strait of Hormuz and there are virtually no alternative supply routes — this uncertainty is not an abstract statistical concept. It translates into electricity bills, manufacturing costs, and ultimately household purchasing power.

5 Conclusion: Supply Routes, Vulnerability, and the Stakes of Diversification

The story of oil price uncertainty under the Iran crisis is ultimately a story about **geography and the absence of alternatives**.

Europe and Asia face the same geopolitical headlines, but they do not face the same economic consequences. The Brent–Dubai spread, which has swung from near-zero to −\$23/barrel in the space of months, tells us that when the Strait of Hormuz comes under threat, Asian economies absorb a shock that simply does not reach European ones with the same force. This asymmetry is not accidental — it is structural, built into decades of energy infrastructure decisions.

For Asia’s most exposed economies — Japan and South Korea — the path forward is supply diversification, and the options are constrained. Japan has been investing in liquefied natural gas (LNG) infrastructure from Australia, the United States, and Canada as a partial substitute for Middle Eastern crude. The US–Japan LNG relationship, strengthened since 2022, offers a genuine long-run alternative — but LNG terminals, regasification capacity, and fleet adaptation take years to build. In the short run, Japan remains trapped: 90% Middle East dependence is not a number that changes in a single policy cycle. A Hormuz closure tomorrow would mean an immediate supply crisis with no available substitute at scale.

South Korea is in a structurally similar position, though its government has moved more aggressively toward nuclear energy as a domestic buffer. Expanding nuclear capacity reduces oil dependence for power generation, but transportation and industrial feedstocks remain tied to crude imports — and therefore to the Strait.

China has the most credible diversification story. The Russia–China pipeline network, expanded since 2022 under Western sanctions pressure on Moscow, gives Beijing a land-based supply route entirely insulated from Hormuz risk. For China, the Iran crisis is a manageable disruption rather than an existential threat. This geopolitical reality helps explain why China has been less vocal than Japan or South Korea in calling for Hormuz security guarantees — it has alternatives, and those alternatives are actively being used.

India occupies a fascinating intermediate position. Its heavy reliance on Middle Eastern crude (65%) creates structural vulnerability, but India has demonstrated remarkable flexibility since 2022 by absorbing large volumes of discounted Russian Urals crude that European buyers abandoned. India’s energy pragmatism — buying from whoever offers the best price regardless of geopolitical alignment — gives it more supply resilience than the headline dependency figure suggests. However, this strategy depends on Russian supply remaining available and cheap, which is itself a geopolitical assumption.

For European economies, the Iran crisis is a risk but not a crisis. Germany and France source the majority of their oil from Norway, the North Sea, Algeria, and increasingly the United States via LNG. The Strait of Hormuz is relevant to European energy security — a full closure would spike global oil prices and hurt European growth — but it would not trigger a supply cutoff. Europe’s challenge is different: the legacy dependence on Russian pipeline gas, partially visible in the “Norway/Russia” alternative route listed for Germany, has already been disrupted by the 2022 war. Europe has been forced into diversification; Asia has not yet been forced, but the current crisis is delivering the same message.

The forecasting models make the stakes concrete. Both benchmarks behave as random walks, confirming that oil prices cannot be reliably predicted — only their uncertainty can be quantified. The Dubai forecast’s extreme upper confidence bound is not a price prediction; it is a measure of how severely the Asian crude market has been destabilised. For Japan, sitting in a country where every barrel of oil travels through a waterway that one nation could close, that uncertainty is not an abstraction. It is the price paid for having built an industrial economy on a foundation of imported energy with no plan B.

The ultimate lesson is that supply diversification is not just an energy policy choice — it is a macroeconomic risk management imperative. The countries that have invested in alternative routes, domestic renewables, or flexible procurement strategies will weather the next geopolitical shock at a lower cost than those that have not. The spread chart in Figure 2 shows who is already paying the price for that difference.

6 Technical Appendix

6.1 Data Sources

Series	Description	Source	Frequency
DCOILBRENTU	Crude Oil Prices: Brent — Europe	U.S. EIA via FRED	Daily
POILDUBUSDM	Global price of Dubai Crude	IMF via FRED	Monthly

Data retrieved via the FRED API (Federal Reserve Bank of St. Louis). Author of analysis and report: Daniela Borja Polanco.

6.2 R Session Information

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## R version: R version 4.4.1 (2024-06-14)

## Key packages: httr, forecast, ggplot2, tidyverse

## Report generated: 08 May 2026
```

6.3 SES and Holt Equations

The Simple Exponential Smoothing level equation:

$$\hat{y}_{t+1|t} = \alpha y_t + (1 - \alpha) \hat{y}_{t|t-1}$$

For both Brent and Dubai: $\alpha = 0.9999$, confirming near-random-walk behaviour.

$$\hat{y}_{t+1} = 0.9999 \cdot y_t + 0.0001 \cdot \hat{y}_t$$

The Holt damped trend model additionally estimates a trend component b_t with damping parameter ϕ :

$$\hat{y}_{t+h|t} = \ell_t + (\phi + \phi^2 + \dots + \phi^h) b_t$$

The damping parameter prevents the forecast from growing unboundedly — particularly important given the extreme recent spike in Dubai crude prices.